Statistics & Probability

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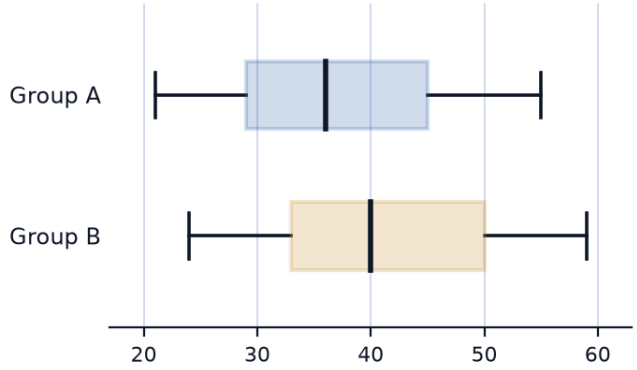
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Mean, median, and mode

Find the mean of $4, 8, 6, 10, 7$.

2 [Foundational] • Target: Mean, median, and mode

Find the median of $3, 9, 5, 1, 7$.



3 [Foundational] • Target: Mean, median, and mode

Find the mode of $2, 4, 4, 6, 8, 4, 2$.

4 [Foundational] • Target: Mean, median, and mode

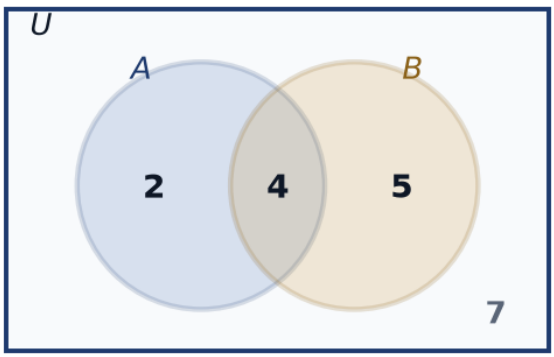
A data set has values $8, 12, 10, 15, 10, 14, 11$. Find the mean, median, and mode.

5 [Foundational] • Target: Range and quartiles

Find the range of $5, 12, 8, 20, 3$.

6 [Foundational] • Target: Range and quartiles

For the data set $2, 4, 5, 7, 8, 9, 12$, find Q_1 and Q_3.



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7 [Foundational] • Target: Range and quartiles

Find the interquartile range (IQR) for the same data set.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Variance and standard deviation

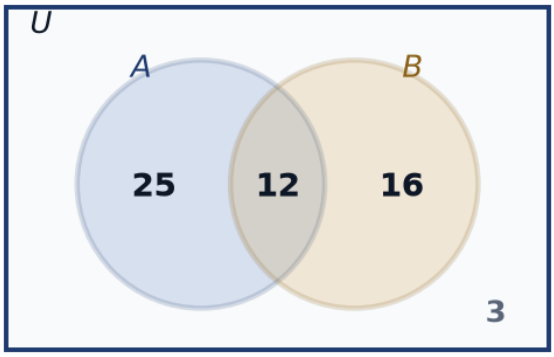
Find the variance of $2, 4, 6, 8, 10$, using $\sigma^2 = \frac{\sum(x-\mu)^2}{n}$.

9 [Intermediate] • Target: Variance and standard deviation

Find the standard deviation of the same data set, rounded to two decimal places.

10 [Intermediate] • Target: Variance and standard deviation

A data set has variance 25. State its standard deviation.



11 [Intermediate] • Target: Frequency tables

A frequency table shows value 1 with frequency 3, value 2 with frequency 5, and value 3 with frequency 2. Find the mean.

12 [Intermediate] • Target: Frequency tables

Using the same frequency table, find the mode.

13 [Intermediate] • Target: Frequency tables

Using the same frequency table, find the median.

14 [Intermediate] • Target: Basic probability

A fair six-sided die is rolled. Find $P(\text{number} > 4)$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

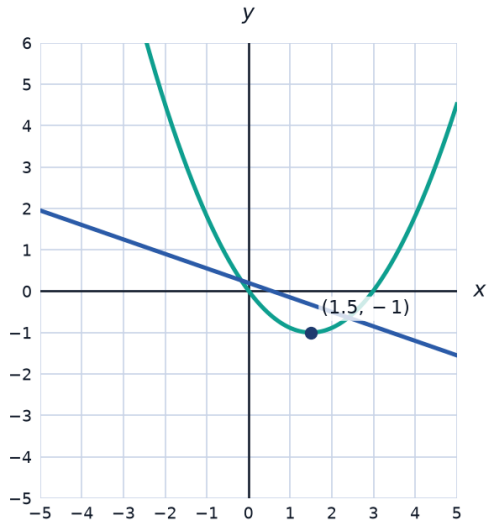
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15 [Challenging] • Target: Basic probability • Cross-Lesson: 7.01 Integration & Applications

A bag contains 5 red and 3 blue marbles. Find $P(\text{blue})$.

16 [Challenging] • Target: Basic probability • Cross-Lesson: 8.01 Exponentials & Logarithms

A card is drawn from a standard 52-card deck. Find $P(\text{king})$.



17 [Challenging] • Target: Basic probability • Cross-Lesson: 1.01 Algebra & Equations

Two fair coins are flipped. Find $P(\text{both heads})$.

18 [Challenging] • Target: Complementary and mutually exclusive events • Cross-Lesson: 2.01 Functions & Transformations

If $P(A) = 0.35$, find $P(\text{not } A)$.

19 [Challenging] • Target: Complementary and mutually exclusive events • Cross-Lesson: 3.01 Sequences & Series

A die is rolled. Events $A = \{\text{even}\}$ and $B = \{\text{rolling a } 5\}$. Are A and B mutually exclusive?

20 [Challenging] • Target: Complementary and mutually exclusive events • Cross-Lesson: 4.01 Trigonometry

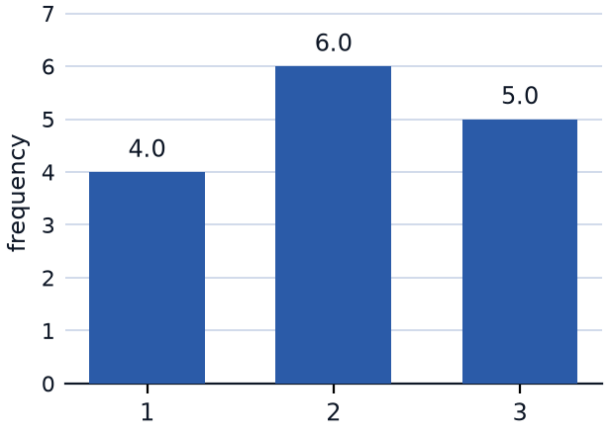
If $P(A) = 0.4$ and $P(B) = 0.25$, and A, B are mutually exclusive, find $P(A \text{ or } B)$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Frequency tables

The bar chart shows a frequency table. Find the total number of data points.



a. 15

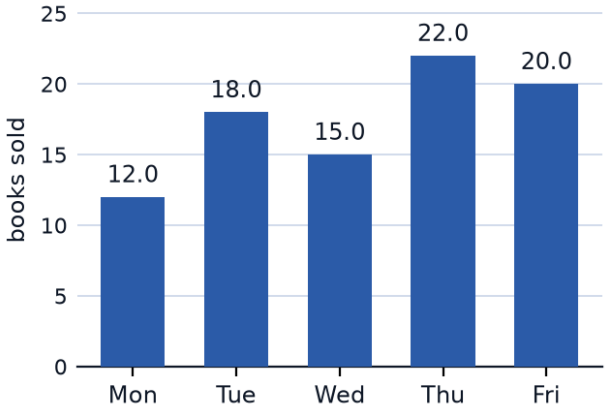
b. 6

c. 3

d. 10

22 [Foundational] • Target: Reading data from a bar chart

The bar chart shows the number of books sold each day at a bookstore over five days. Find the total books sold.



a. 87

b. 22

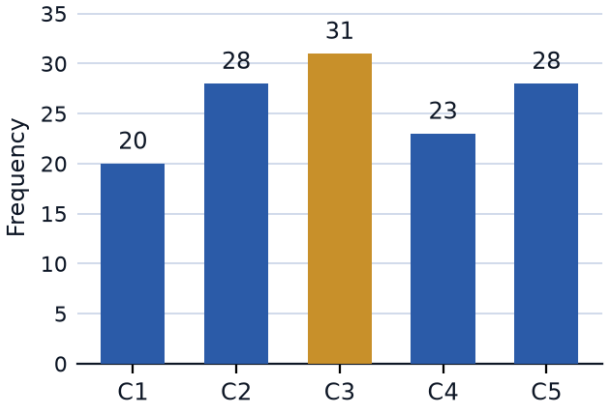
c. 77

d. 97

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23 [Foundational] • Target: Reading data from a bar chart

Using the same bar chart, find the mean number of books sold per day.



a. 15

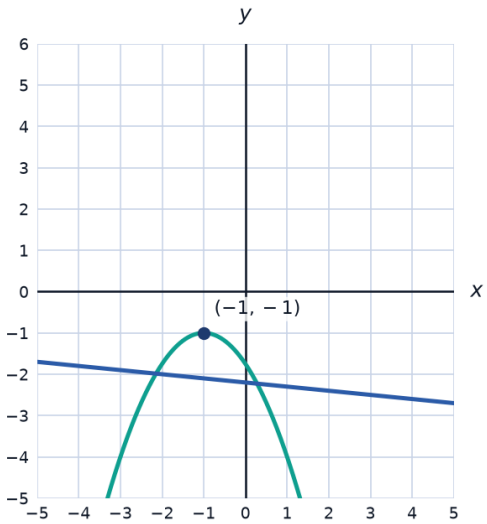
b. 18

c. 20

d. 17.4

24 [Foundational] • Target: Reading data from a bar chart

Using the same bar chart, find the range of books sold.



a. 12

b. 22

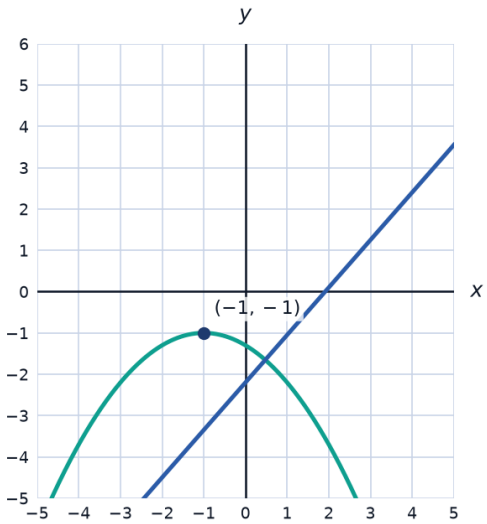
c. 8

d. 10

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25 [Foundational] • Target: Reading data from a bar chart

Using the same bar chart, on which day were the most books sold?



a. Wednesday

b. Thursday

c. Tuesday

d. Friday

26 [Foundational] • Target: Frequency tables

A frequency table shows value 1 with frequency 4, value 2 with frequency 6, and value 3 with frequency 5. Find the total number of data points.

a. 10

b. 15

c. 3

d. 6

27 [Foundational] • Target: Frequency tables

Using the same frequency table, find the mean.

a. 2.33

b. ≈ 2.07

c. 2.00

d. 1.87

28 [Foundational] • Target: Basic probability

A fair six-sided die is rolled. Find $P(\text{number} < 3)$.

a. $\frac{1}{6}$

b. $\frac{1}{2}$

c. $\frac{1}{3}$

d. $\frac{2}{3}$

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Basic probability**

A bag contains 7 red and 4 blue marbles. Find $P(\text{red})$.

a. $\frac{7}{11}$ **b.** $\frac{1}{11}$

c. $\frac{4}{11}$ **d.** $\frac{7}{4}$

30 **[Intermediate]** • **Target: Basic probability**

A card is drawn from a standard 52-card deck. Find $P(\text{queen})$.

a. $\frac{4}{13}$ **b.** $\frac{1}{4}$

c. $\frac{1}{13}$ **d.** $\frac{1}{52}$

31 **[Intermediate]** • **Target: Basic probability**

Two fair coins are flipped. Find $P(\text{at least one tail})$.

a. $\frac{1}{2}$ **b.** $\frac{3}{4}$

c. $\frac{1}{4}$ **d.** $\frac{1}{3}$

32 **[Intermediate]** • **Target: Basic probability**

A fair six-sided die is rolled. Find $P(\text{multiple of } 3)$.

a. $\frac{2}{5}$ **b.** $\frac{1}{3}$

c. $\frac{1}{2}$ **d.** $\frac{1}{6}$

33 **[Intermediate]** • **Target: Complementary and mutually exclusive events**

A die is rolled. Events $A = \{\text{odd}\}$ and $B = \{\text{rolling a } 4\}$. Are A and B mutually exclusive?

a. Yes **b.** Only sometimes

c. No **d.** Cannot be determined

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34 [Intermediate] • Target: Independent events

A fair coin is flipped twice. Find $P(\text{both heads})$.

a. $\frac{1}{3}$

b. 1

c. $\frac{1}{4}$

d. $\frac{1}{2}$

35 [Intermediate] • Target: Independent events

A card is drawn from a deck, replaced, then a second card is drawn. Find $P(\text{both are kings})$.

a. $\frac{2}{13}$

b. $\frac{1}{26}$

c. $\frac{1}{13}$

d. $\frac{1}{169}$

36 [Intermediate] • Target: Conditional probability

In a class, $P(\text{science}) = 0.65$ and $P(\text{math and science}) = 0.3$. Find $P(\text{math} \mid \text{science})$.

a. 0.65

b. 0.195

c. ≈ 0.46

d. 0.3

37 [Intermediate] • Target: Conditional probability

A box has 12 balls: 7 red and 5 blue. Two balls are drawn without replacement. Given the first is blue, find $P(\text{second is blue})$.

a. $\frac{5}{11}$

b. $\frac{4}{11}$

c. $\frac{5}{12}$

d. $\frac{4}{12}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Conditional probability • Cross-Lesson: 2.01 Functions & Transformations

Given $P(A) = 0.45$, $P(B) = 0.5$, $P(A \text{ and } B) = 0.2$, find $P(B \mid A)$.

a. 0.4

b. 0.09

c. 0.5

d. ≈ 0.44

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39 [Challenging] • Target: Counting: permutations and combinations • Cross-Lesson: 3.01 Sequences & Series

In how many ways can 6 people be arranged in a line?

a. 720

b. 120

c. 36

d. 6

40 [Challenging] • Target: The five-number summary • Cross-Lesson: 4.01 Trigonometry

Outlier boundaries are $Q_1 - 1.5 \times IQR$ and $Q_3 + 1.5 \times IQR$. Find both boundaries for the same data set.

a. -15 and 49

b. 9 and 25

c. -15 and 40

d. -24 and 40

41 [Challenging] • Target: The five-number summary • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

For the data set $10, 15, 20, 25, 30, 35, 40$, find the median.

a. 25

b. 30

c. 20

d. 27.5

42 [Challenging] • Target: Word problems: statistics and probability in context • Cross-Lesson: 6.01 Limits & Derivatives

A survey of 250 students found that 150 like math and 100 like science, with 40 liking both. Find the probability that a randomly selected student likes math or science.

a. 0.84

b. 0.6

c. 0.16

d. 1.0

43 [Challenging] • Target: Word problems: statistics and probability in context • Cross-Lesson: 7.01 Integration & Applications

A quality-control inspector finds that 3% of items from a production line are defective. If 500 items are inspected, how many are expected to be defective?

a. 150

b. 15

c. 3

d. 30

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44 [Challenging] • Target: Word problems: statistics and probability in context • Cross-Lesson: 8.01 Exponentials & Logarithms

In a survey of 180 people, 108 prefer coffee over tea. Find the probability, as a percent, that a randomly selected person prefers coffee.

a. 60%

b. 50%

c. 108%

d. 40%

45 [Challenging] • Target: Statistics & Probability • Cross-Lesson: 1.01 Algebra & Equations

Two fair coins are flipped. Find $P(\text{one head one tail})$.

a. $\frac{1}{4}$

b. $\frac{1}{2}$

c. $\frac{3}{4}$

d. $\frac{1}{8}$

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